Actionable Real-Time Modeling of Surgical Team Dynamics via Time-Expanded Interaction Graphs

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Operating Rooms Design

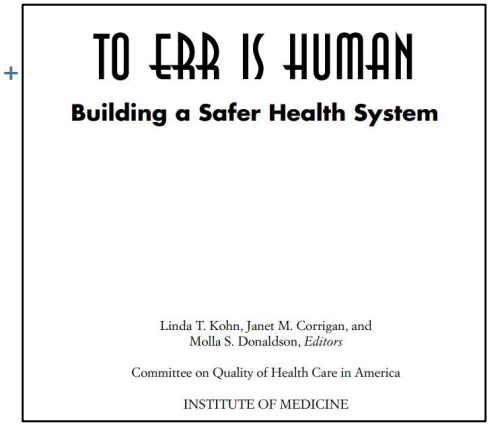


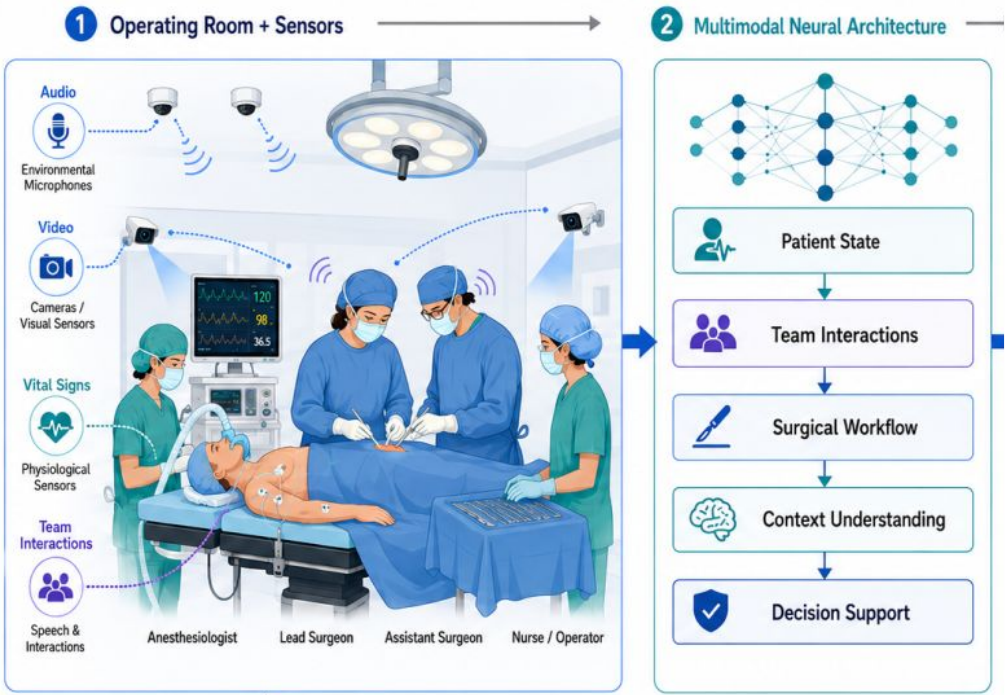
Figure: (Brown, 2001)

+ Medical errors are the **third leading cause of death** in the US
+ One of the main reasons for medical errors is **poor teamwork in the OR**
+ Lack of datasets and annotations capturing interactions in the OR
+ Lack of architectures modeling team behavior generating accurate predictions
+ Lack of systems able to provide feedback

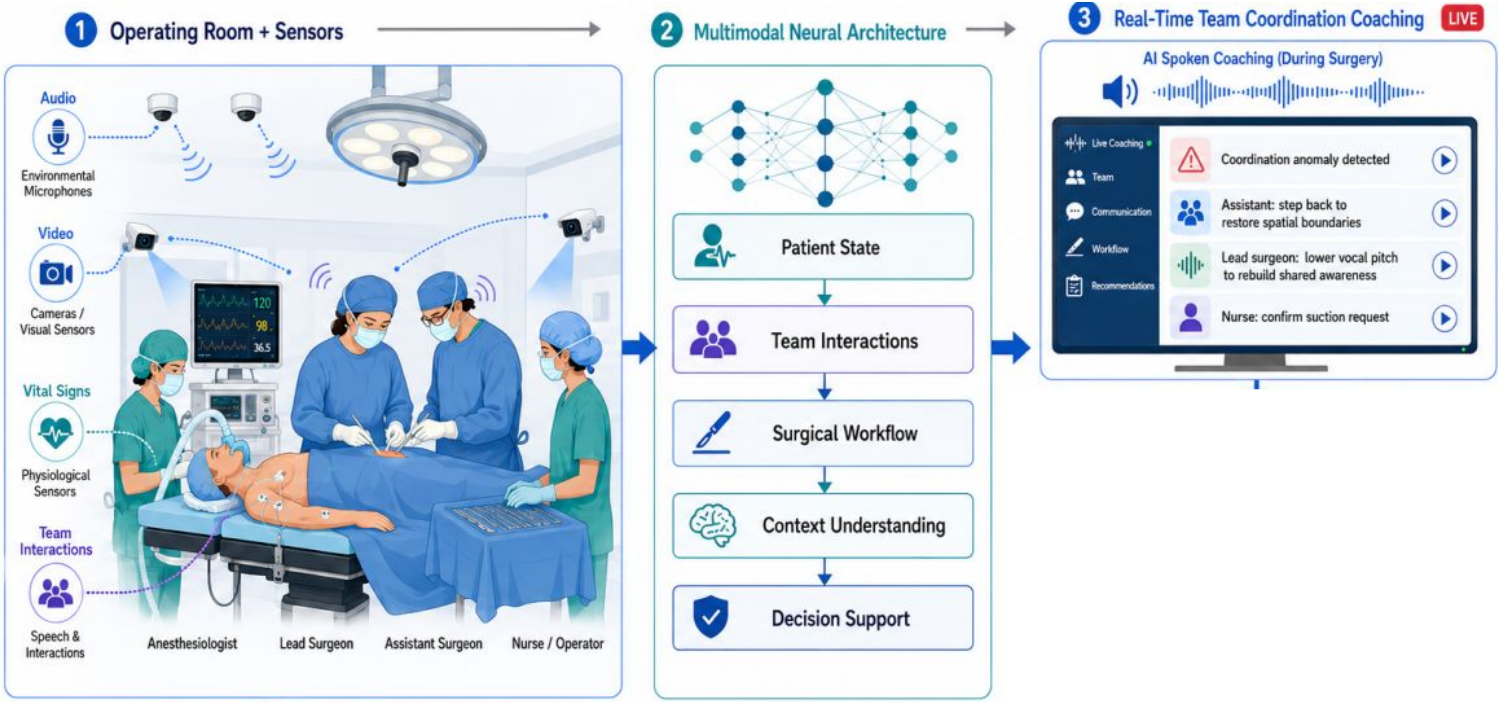
Operating Rooms Design



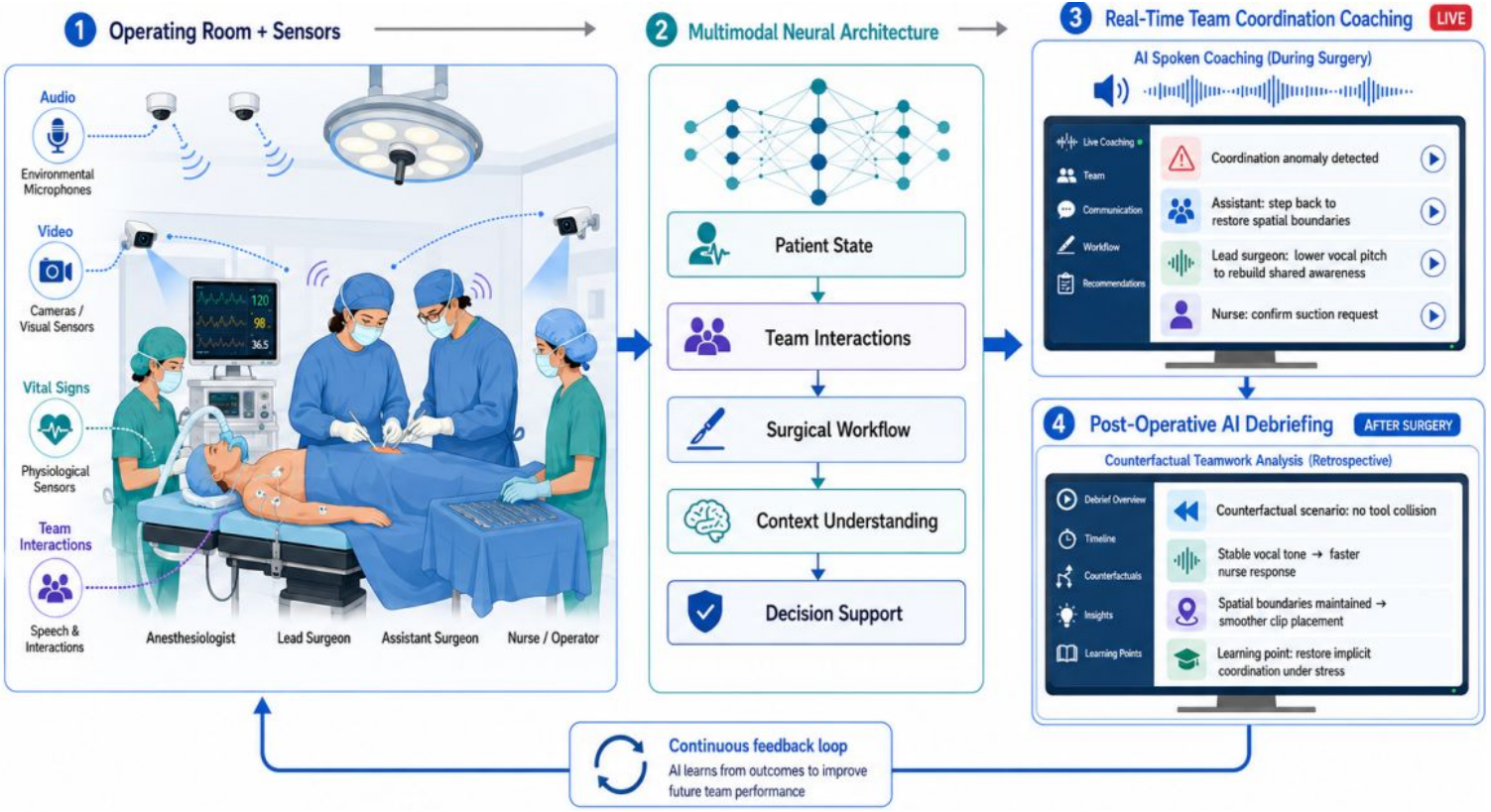
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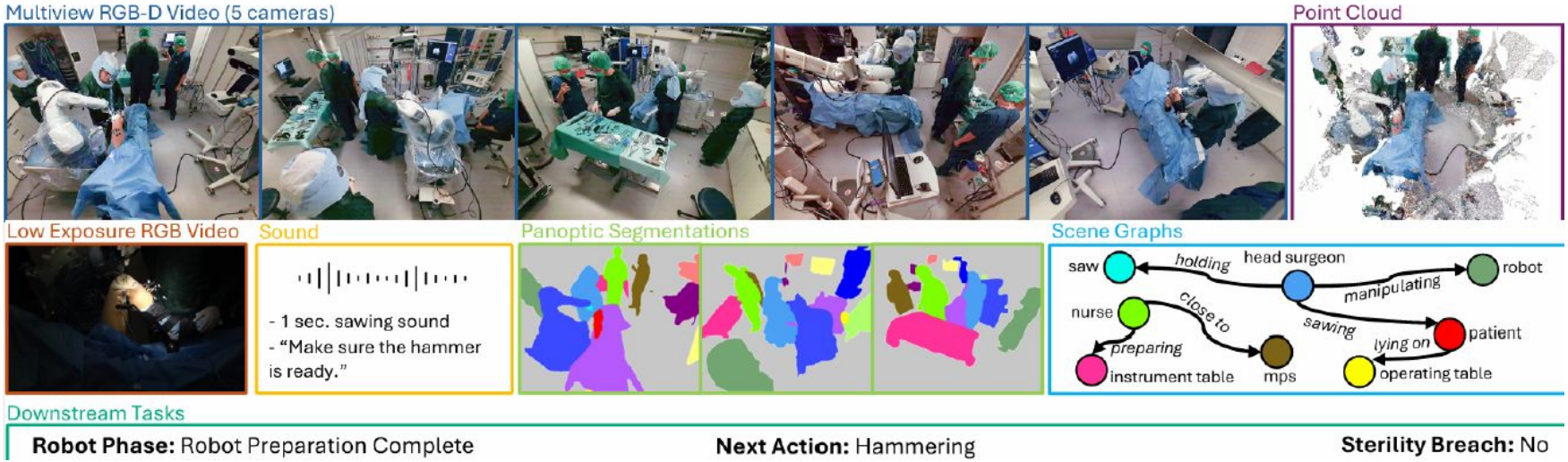
Operating Rooms Design



Operating Rooms Design



Operating Room Data

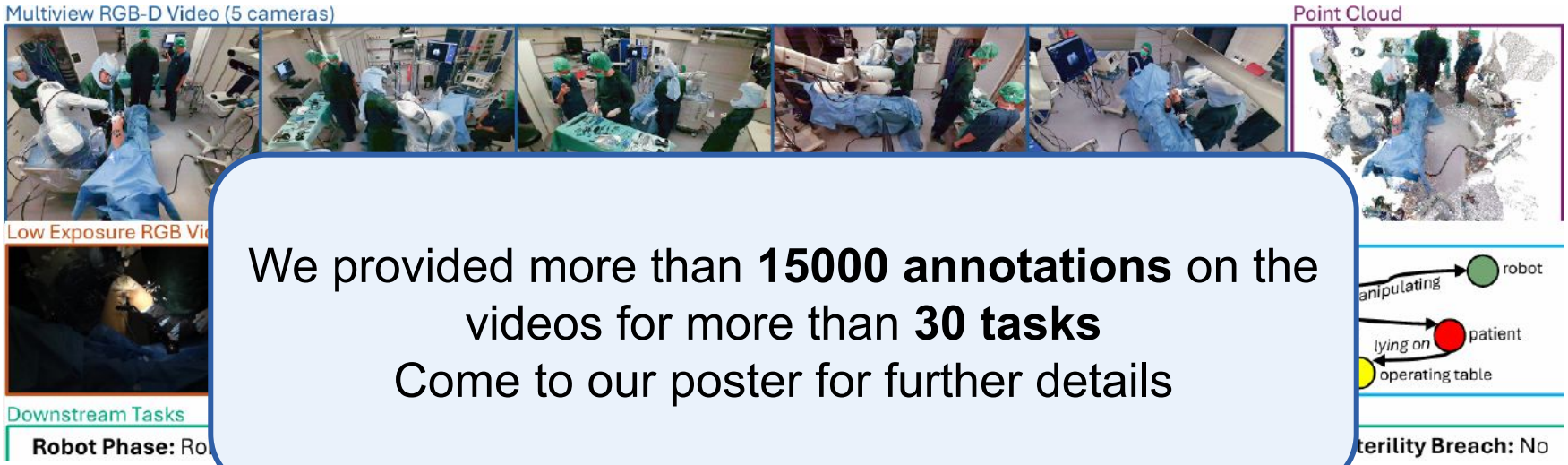


+ MM-OR: 15 hours of multimodal data collected in Operating Room

+ Lack of proper **multimodal feature extraction**

+ Lack of proper **data labeling** for social interactions

Operating Room Data

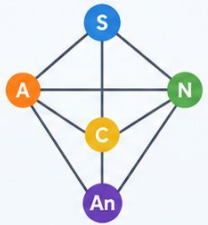


We provided more than **15000 annotations** on the videos for more than **30 tasks**
Come to our poster for further details

+ MM-OR: 15 hours of multimodal data collected in Operating Room
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Model Operating Room

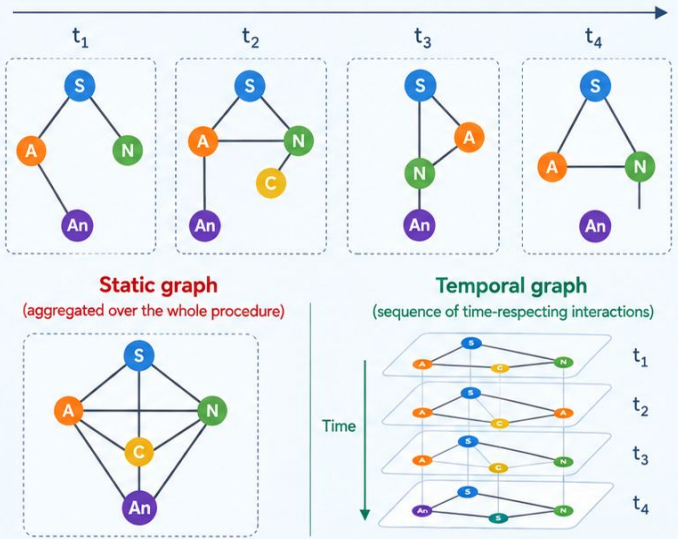
Static Graph



- Treats interactions as persistent
- Loses event order
- Misses transient collaborations

Temporal Graph

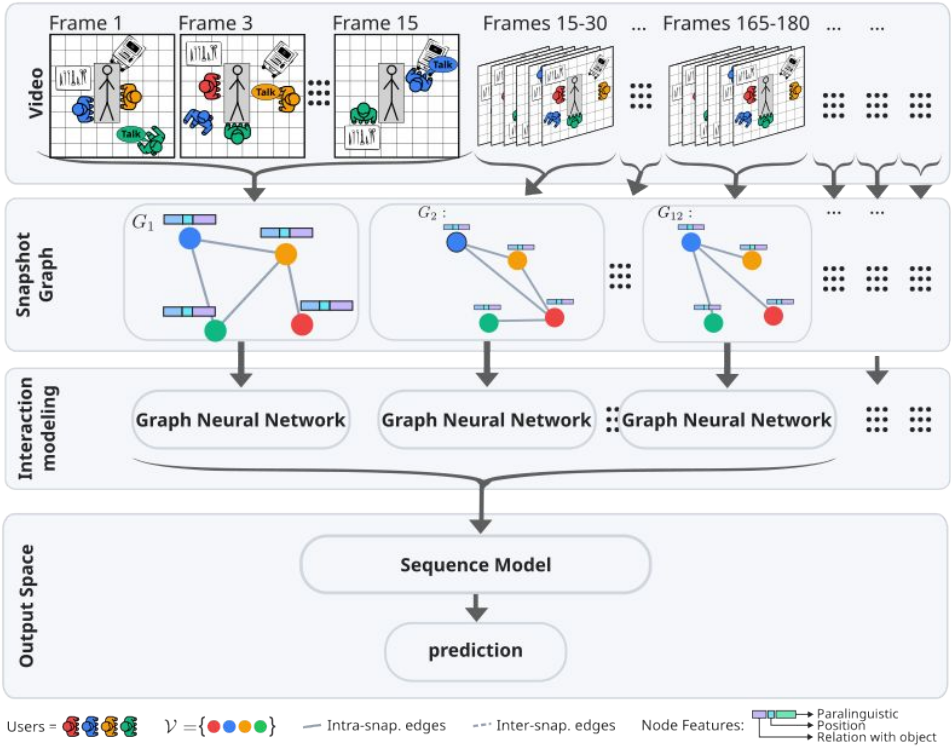
- Captures when interactions occur
- Preserves interaction sequence
- Reveals changing team dynamics



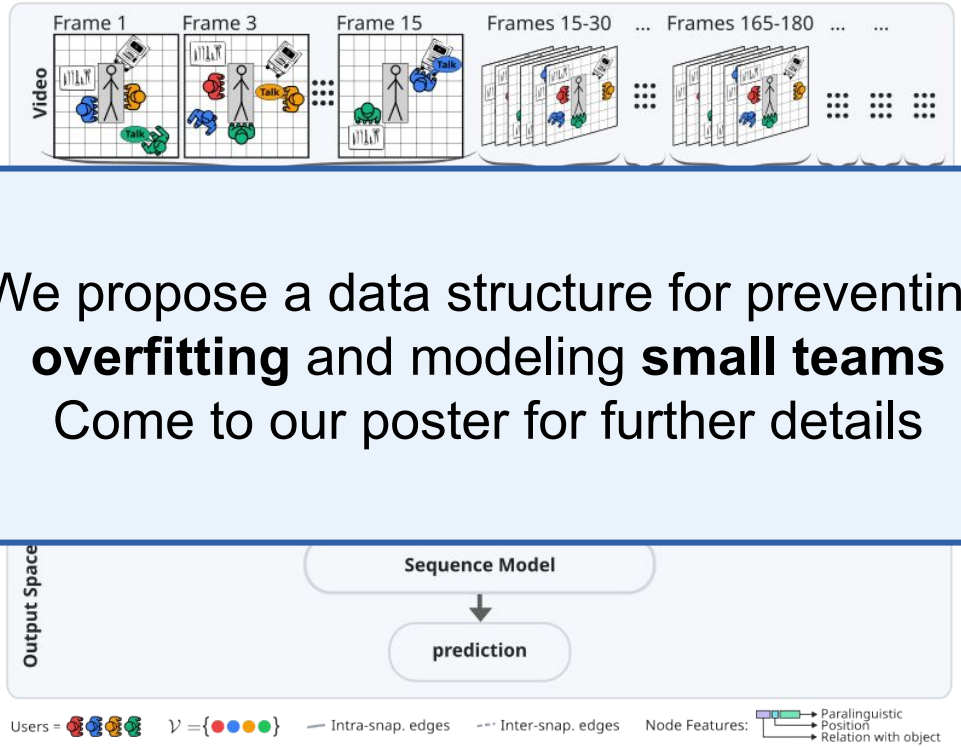
Challenge

Small teams • Limited temporal data

Temporal Graph Neural Networks



Temporal Graph Neural Networks



Explanations

FACTUAL EXPLAINABILITY

Why did the AI give this prediction?



AI PREDICTION:
HIGH RISK
of conflict in the surgical team

WHY?

Higher-than-expected number of surgeons on standby	+0.42
Recent tense communication in the last 24 hours	+0.31
Surgery duration expected to be longer than usual	+0.21

The model flagged this risk because of these observable **team conditions and behaviors** (e.g., more people, recent tension, longer case).

VS

COUNTERFACTUAL EXPLAINABILITY

What could change the prediction?



AI PREDICTION:
LOW RISK
of conflict in the surgical team

WHAT COULD CHANGE THE PREDICTION?

If the following changes:

↓ Reduce number of surgeons on standby (e.g., from 5 to 2)	→	RISK DECREASES **LOW**
↓ Resolve recent tension (e.g., through structured brief)	→	RISK DECREASES **LOW**
↓ Shorten expected surgery duration (e.g., optimize schedule)	→	RISK DECREASES **LOW**

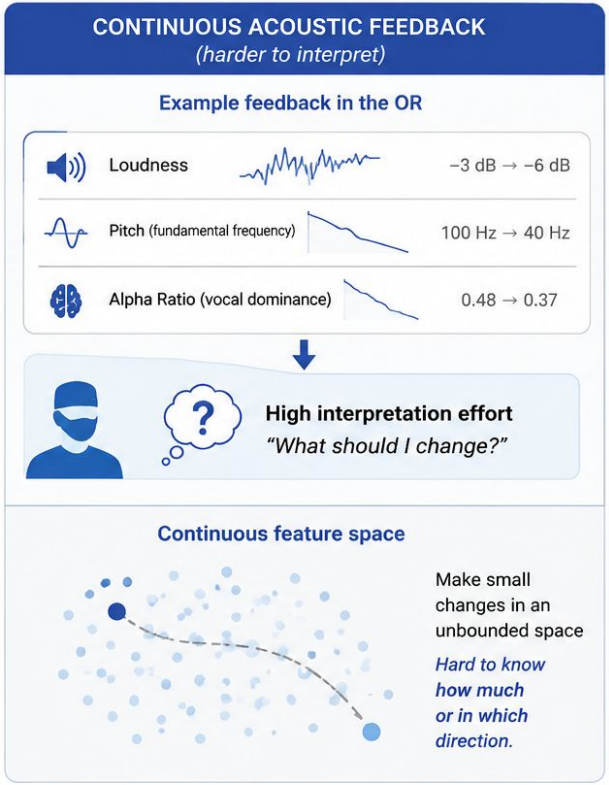
If these **team behaviors** change, the likelihood of conflict drops significantly.

IN SHORT

Factual = explains the current team situation (e.g., more people, tension, longer case).
Counterfactual = suggests what could be changed (e.g., adjust staffing, improve communication, shorten time) to reduce the risk of conflict.



Interpretable explanations



Interpretable explanations

CONTINUOUS ACOUSTIC FEEDBACK
(harder to interpret)

Example feedback in the OR

We propose an algorithm that generate **interpretable** and **plausible** feedback.
Come to our poster for further details

Continuous feature space



Make small changes in an unbounded space

Hard to know how much or in which direction.

Feedback example

1) Adding Communication



Situation. The nurse asks a question, but the leader is distracted and does not respond, leading to silence and low communication.

Counterfactual. Add edges

Interpretation. Reduce disengagement by increasing leader responsiveness.

4) Aggressive Communication

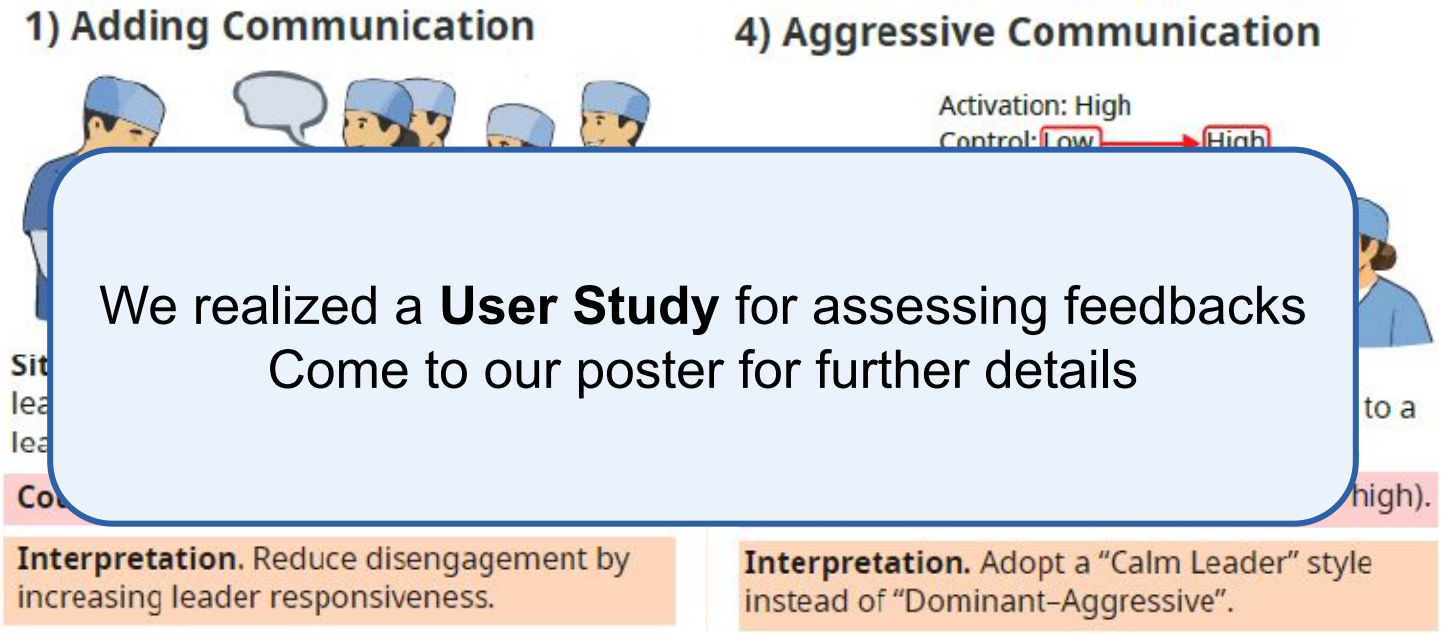


Situation. A rude response from the leader to a nurse's question reduces team cooperation.

Counterfactual. Increase "Control" (low → high).

Interpretation. Adopt a "Calm Leader" style instead of "Dominant–Aggressive".

Feedback example



We realized a **User Study** for assessing feedbacks
Come to our poster for further details

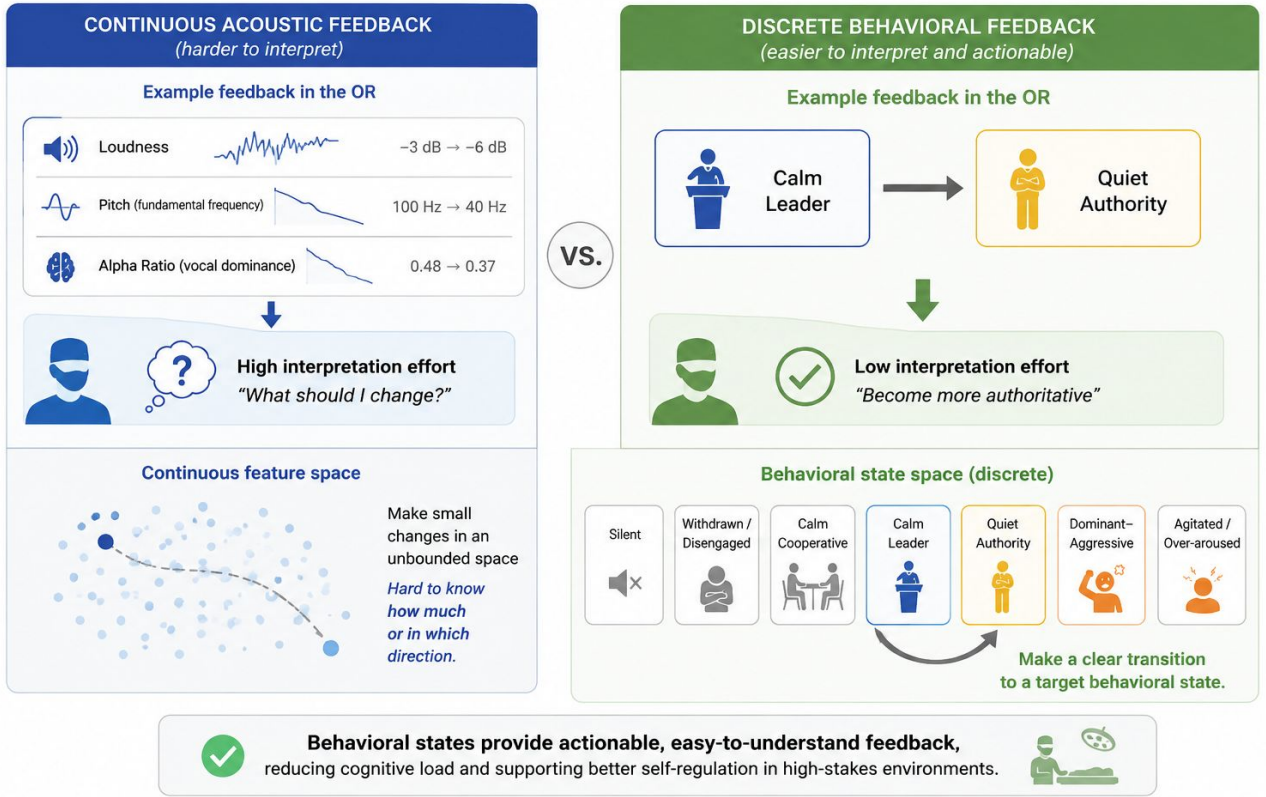
Thank you for the attention!

Appendix - Behavioral classes

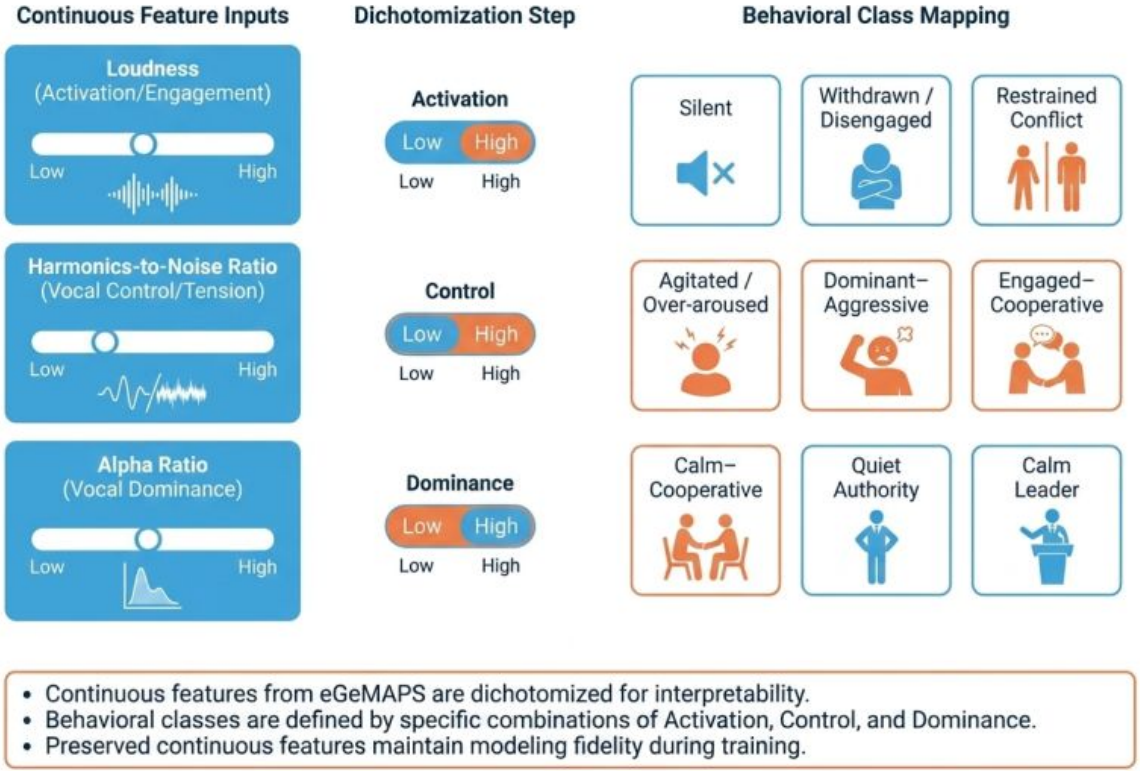
Activation	Control	Dominance	Behavioral Class
Low	Low	Low	Withdrawn / Disengaged
Low	Low	High	Restrained Conflict
Low	High	Low	Calm–Cooperative
Low	High	High	Quiet Authority
High	Low	Low	Agitated / Over-aroused
High	Low	High	Dominant–Aggressive
High	High	Low	Engaged–Cooperative
High	High	High	Calm Leader

Table 1: Behavioral classes defined by combining three paralinguistic features extracted from the eGeMAPS set.

Interpretable feedback



Interpretable feedback



- Continuous features from eGeMAPS are dichotomized for interpretability.
- Behavioral classes are defined by specific combinations of Activation, Control, and Dominance.
- Preserved continuous features maintain modeling fidelity during training.